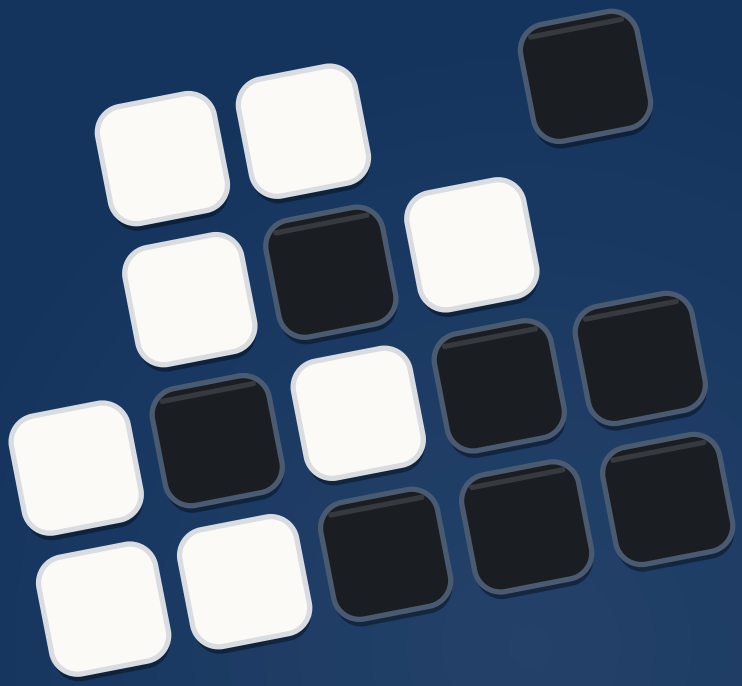
A GAME BY GUILLAUME ENGUEHARD

TSUGI

Tactics and strategy



Sixteen pieces on the table.
One group.
Seven to connect.

2 PLAYERS · NO BOARD

INTRODUCTION

Understanding Tsugi

This guide gathers what you need to play well: the ways to win and to defend, combinations, endgames and strategic principles. It is based on 300 games played by a computer that looks 6 moves ahead. Every pattern was detected and checked by program, and every diagram is a position that actually occurred.

GETTING STARTED

Five habits

1. **Look at your opponent first.** Before every move, look for their winning moves: junction, stray piece, smothering. Then check that your move does not offer one.
2. **Multiply contacts.** Keep your pieces side by side, in small groups close to one another.
3. **Connect at the last moment.** A group of 6 and a piece ready to join it are worth more than a large group shown too early.
4. **Pin.** Place your pieces so that enemy pieces cannot move without splitting the common group.
5. **Count the moves.** A player who cannot move loses: keep track of how many moves each side has left.

CONTENTS

1 **Winning in one move**
Junction, stray piece, smothering

2 **Defending**
The six possible parries

3 **Three-move combinations**
Zugzwang, forbidden parry, fork...

4 **Long combinations**
The quiet move

5 **Endgames**
Winning shapes, the eighth piece, blockade

6 **Strategy**
What foretells victory, nine principles

7 **Course of a game and openings**
The phases of a game, the first moves

READING THE DIAGRAMS

piece to move

move played and landing square

winning threat

opponent's move

forbidden move

winning group

GLOSSARY

Stepping stone	A piece you jump over. The opponent's serve as well as your own.
End of line	The empty square where a slide stops.
Threat	A move that would win next turn if the opponent does nothing.
Parry	A move that makes a threat fail.
Pinned piece	A piece that cannot move without splitting the common group of 16 pieces.
Quiet move	A move with no immediate threat that prepares a combination.
Zugzwang	A position where every move loses for the player who must move.
Move	One player's turn (a “ply”).

The families of wins and parries are complete: every forced win and every parry falls into one of them. The frequencies depend on the computer's style; expert human players will no doubt bring out other plans. The strategic principles are correlations: they say what goes with victory, not why.

CHAPTER 1

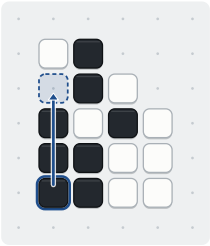
Winning in one move

Every one-move win belongs to one of these three families. Spotting them at a glance, on your side as on your opponent's, is the first skill of the game.

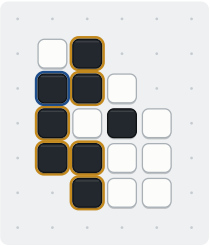
A1 Junction

37% of one-move wins

A piece jumps over its own pieces and welds two groups separated by a single square. To prepare it, build two nearby groups totalling 7 pieces, with one piece able to cross one group and land against the other.



Black slides this piece over its own pieces.



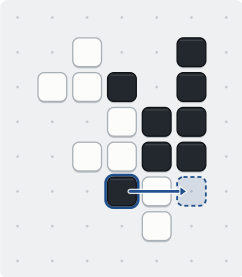
The two groups become one: 7 pieces connected.

A2 Bringing in a stray piece

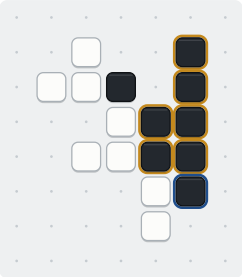
41%

An isolated piece joins a group of 6, using the opponent's pieces as stepping stones. This is the most common win: a group of 6 is always dangerous, even when the seventh piece is far away.

Winning slides jump one piece in 35% of cases, two in 40%, three or more in 25%: read the long lines.



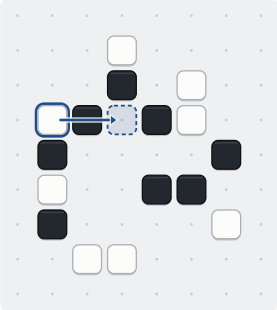
The isolated black piece jumps over the white pieces.



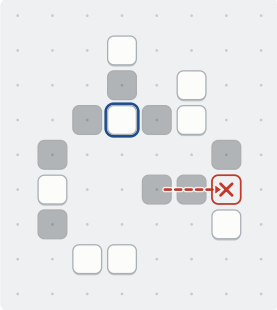
It completes a group of 7.

A3 Smothering 22%

You win without connecting anything: after your move the opponent has no legal move left, and a player who cannot move loses. Before every move, count the moves your opponent will have left.



White plays this move.



Black has no move left and loses. The crossed-out slide is forbidden by “no going back”.

CHAPTER 2

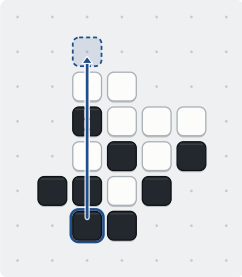
Defending

When the opponent threatens to win next move, there are only six ways to parry, and this list is complete. In the games studied, one threat in three already had no parry and one in four had a single one: you must see threats before they are made.

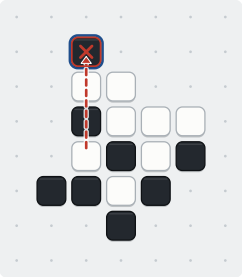
D1 Move the end of the line

37% of parries

Occupy the landing square, or lengthen or shorten the line being jumped: the enemy piece stops elsewhere and connects nothing.



White threatens to win (red dashes). Black parries.

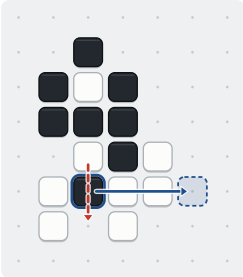


The threat no longer works.

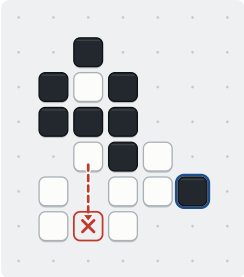
D2 Remove the stepping stone

23%

Move the piece the opponent wanted to jump: with no neighbour in that direction, it can no longer slide.



White threatens to win. Black parries.

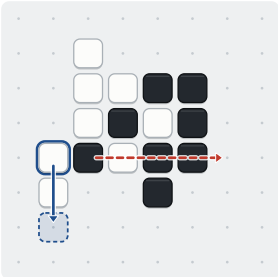


The stepping stone is gone.

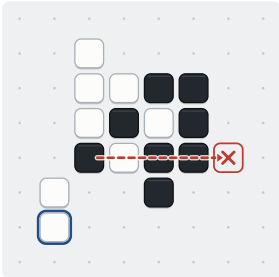
D3 Pin the attacking piece

23%

Play so that the opponent's slide would split the common group of 16 pieces: it becomes illegal.



Black threatens to win. White parries.

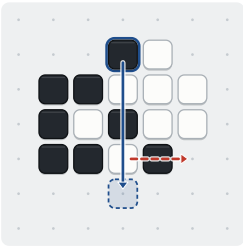


Black's slide would now split the group: forbidden.

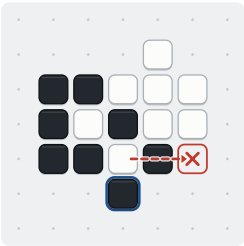
D4 Enclose the attacking piece

14%

Fill the last free square around the attacking piece: once enclosed, it cannot move. A rare variant (1%): turn that last open side into a desert.



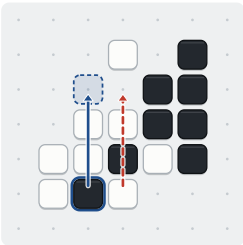
White threatens to win. Black parries.



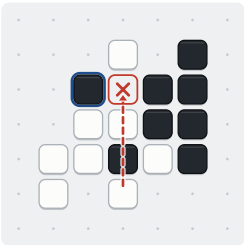
The white piece is enclosed.

D5 Create a desert 3%

Close the fourth side of the landing square: it becomes a desert, where no piece may stop.



White threatens to win. Black parries.



The landing square has become a desert.

NO COUNTER-ATTACK

A counter-threat never saves you: if your opponent can win next move, they win first. Only the parries in this chapter count.

WATCH EVERY FREE PIECE

A group of 6 plus one piece able to join it is all your opponent needs. One of their pieces being enclosed or pinned does not protect you: they can leave it behind and win with the other 7.

CHAPTER 3

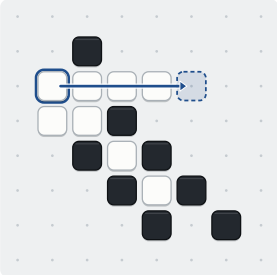
Three-move combinations

A move, a forced reply, then the winning move. The first move belongs to one of these six families. The most common is also the quietest: a move that threatens nothing.

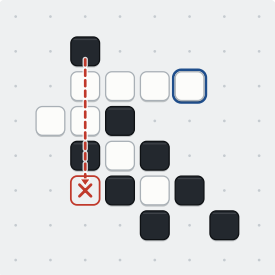
B1 Zugzwang

24% of wins in 3

The move threatens nothing, but the opponent must move, and every one of their moves opens a winning line. Since a blocked player loses, the obligation to move is a first-class weapon.



White plays a quiet move.

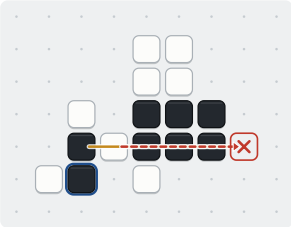


Black has 10 moves, and each lets White win. The crossed-out slide is forbidden by “no going back”.

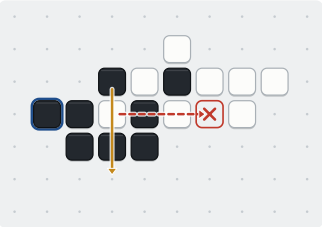
B2 Forbidden parry

21%

A parry exists on the table, but the rules forbid it. Most often the enemy piece serving as a stepping stone is pinned: removing it would split the common group. Or it is enclosed, or its parry would take it back to the square it just left.



Pin: removing the stepping stone would stop the golden threat, but would split the group.

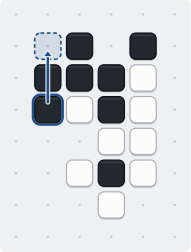


Enclosure: the white piece serving as a stepping stone cannot step aside.

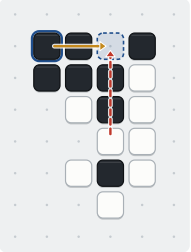
B3 A parry that opens another line

21%

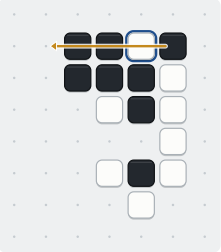
The opponent can parry, but every parry moves a piece that was closing another line: by parrying, they create a new winning threat themselves.



Black plays this move...



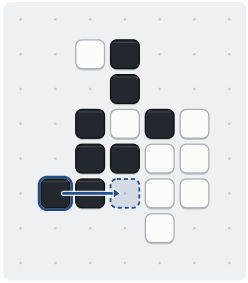
...and threatens to win. White parries...



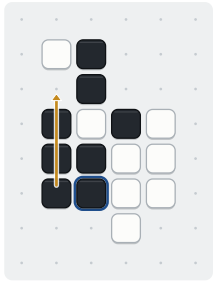
...but opens a new winning line.

B4 Unstoppable threat 19%

No enemy slide removes the stepping stone, occupies the landing square, moves the end of the line or prevents the group of 7 from forming.



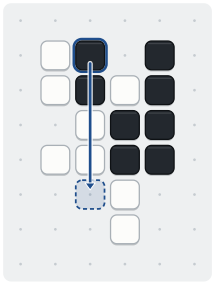
Black plays this move...



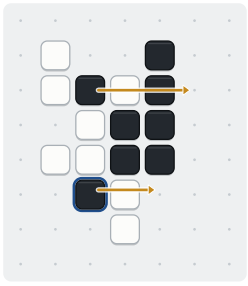
...no white move stops the golden threat.

B5 Fork 10%

A single move creates two winning threats, sometimes three. The opponent can parry only one. Good defenders see it coming, hence its relative rarity.



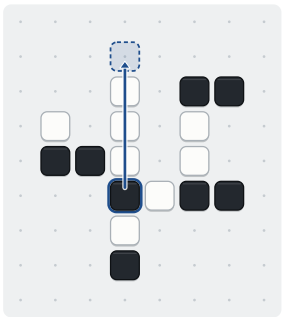
Black plays this move...



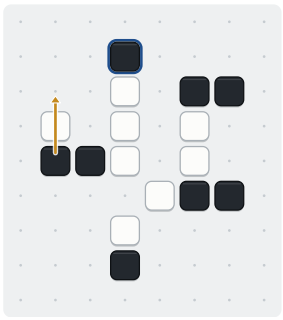
...and threatens to win in two ways.

B6 Smothering threat 5%

The move threatens to leave the opponent with no move at all next turn, and no reply prevents it.



Black plays this move...



...and threatens to leave White with no move (golden arrow).

THE EXACT MOVE

In 69% of positions won in three moves, only one move wins. Combinations do not forgive approximations.

CHAPTER 4

Long combinations

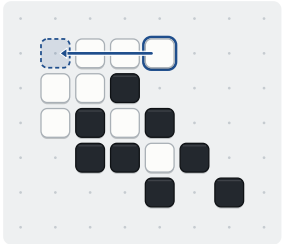
Between players who see short threats, the game is decided by a longer combination the loser did not see. These combinations almost always begin in silence.

C1 The quiet move

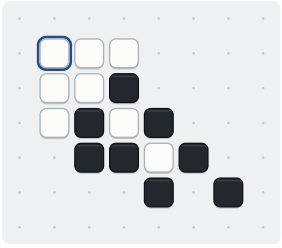
79% of wins in 5, 88% in 7

The first move threatens nothing: it sets up a stepping stone, an end of line or a pin whose threats will become unstoppable two or three moves later. This is what makes the game hard to read.

Other, rarer openings: a series of single threats that wear down the opponent's position (17% of wins in 5), or a fork prepared from afar (2%).



White plays a move with no immediate threat...



...but the position is already a forced win in 5 moves.

HOW FAR AHEAD?

Forced wins in 9 moves are as frequent as those in 5 or 7: about one position in seventy in the games studied, at each depth. The game rewards deep calculation, and a player who can see five moves ahead will put it to use.

CHAPTER 5

Endgames

In the games studied, 68% end with a connection, 20% with a blockade and 12% in a draw. These are two quite distinct ways to win.

E1 The shape of the winning group 76 shapes in 204 wins

There is no typical shape: the most frequent accounts for only 6% of wins, and none is a straight line. Winning groups are compact: two in three fit in a 3 × 4 or 4 × 4 rectangle. Here are the eight most frequent shapes.



12 wins



10 wins



9 wins



8 wins



7 wins



7 wins



6 wins



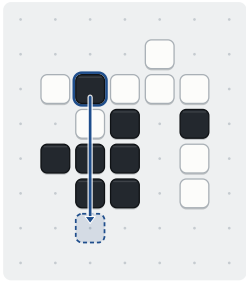
6 wins

E2 The eighth piece

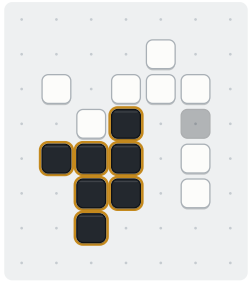
94% of wins

The eighth piece almost always stays outside the winning group: it touches it at a corner in 47% of cases and is one square away in 40%.

It is often a prisoner: pinned in 29% of cases, enclosed in 14%. A blocked piece is not a permanent handicap: you can sacrifice it.



Black plays the winning move.



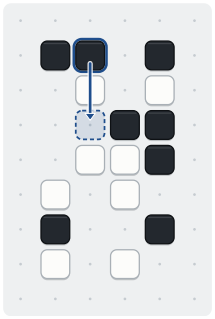
7 pieces connected. The eighth, greyed, is pinned: the winner simply left it behind.

E3 Winning by blockade

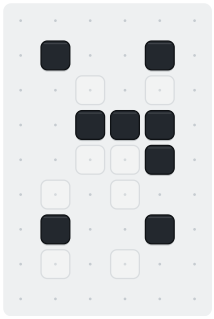
22% of wins

Blocked pieces are blocked mostly because every slide would split the common group (67%) or because they have nothing to jump (24%), sometimes because their only move would take them back (7%). Almost never by enclosure.

At the moment of the blockade, the winner's largest group often has only 2 or 3 pieces: you win by stretching the mass into diagonal chains and pinning, not by grouping up.



Black moves.

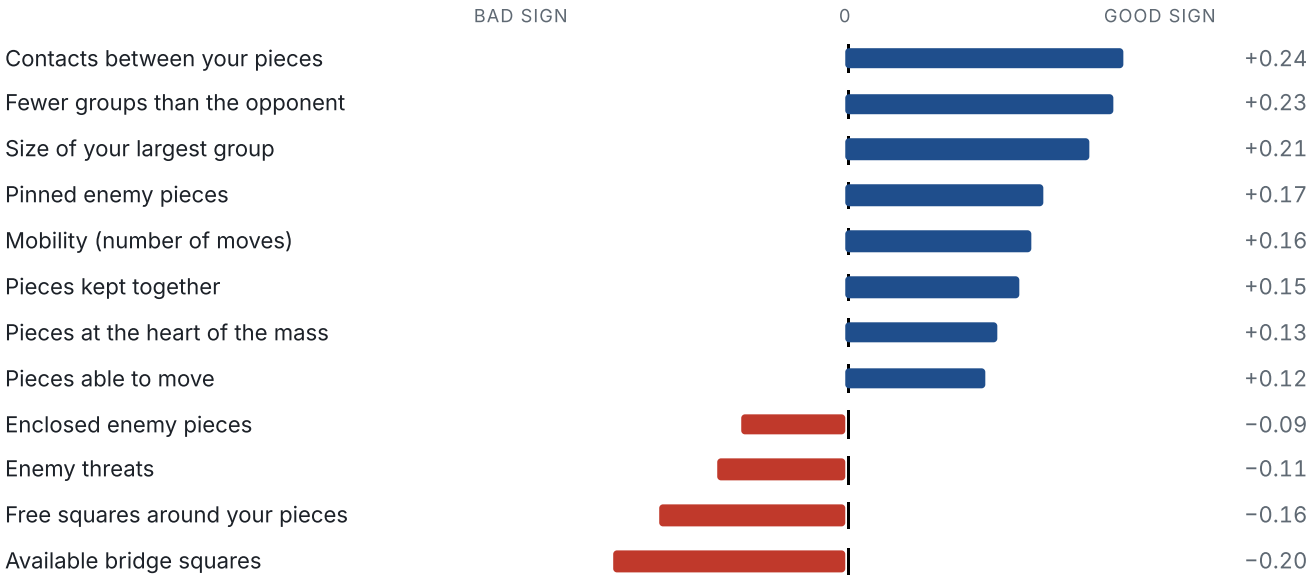


No white piece can move: White loses.

CHAPTER 6

Strategy

What, in a position, foretells victory. The chart measures the link between each criterion, taken alone, and the victory of the player to move, over 19,226 positions. The nine principles take into account the effect of each criterion with all others held equal.



1

Multiply contacts

The number of side-to-side contacts between your own pieces is the best single predictor of victory. It becomes decisive in the endgame.

2

Gather, connect late

Stay less scattered than your opponent, but do not connect early: right to the end, each side keeps 4 to 5 groups of 3 or 4 pieces. Build a network of small, close groups and join them at the last moment.

3

Immobilise the opponent

Every pinned or enclosed enemy piece is a clear advantage. But enclosing often ties up your own pieces: it only pays if you stay compact.

4

Count movable pieces

Having more pieces able to move is an advantage. The raw number of moves says nothing: often they are isolated pieces sliding into empty space.

5

The centre does not matter

Holding the heart of the mass brings nothing by itself. What matters is how your pieces touch.

6

Leave no threat standing

Every pending enemy threat is a bad sign, even a parryable one: it forces a parry, often a single one.

7

Keep the blockade in mind

A second way to win and a constant danger. Count each side's moves. Blocking positions are made of diagonal chains and pinned pieces.

8

Be ready to abandon a piece

A pinned or enclosed piece can stay behind: you win with the other 7. Blocking an enemy piece is not enough to keep you safe.

9

Shift priorities

Opening: everything is still open. Middlegame: contacts and number of groups. Endgame: contacts, gathering and mobility; bridge squares then betray a side still split into pieces.

CHAPTER 7

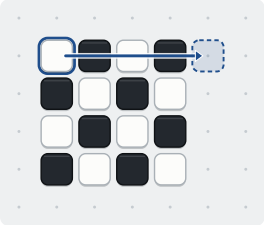
Course of a game and openings

Averages per side over 300 games. As the game goes on, enclosures disappear and pins replace them: the mass stretches out, the central pieces are freed, and it is the cohesion of the common group that immobilises. From move 20 on, about one position in twenty contains a threat to parry.

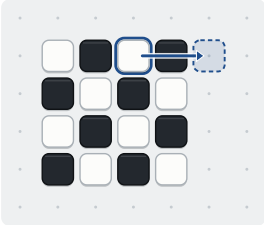
MOVES	GROUPS	LARGEST GROUP	ENCLOSED PIECES	PINNED PIECES	LEGAL MOVES	UNDER THREAT
0–9	7.7	1.3	2.1	0.0	12.6	0%
10–19	5.6	3.2	2.0	0.4	11.0	0%
20–29	4.9	3.7	1.7	0.7	10.6	4%
30–39	4.8	3.7	1.5	0.9	10.3	3%
40–59	4.6	3.8	1.4	1.0	10.1	5%
60–79	4.5	3.9	1.2	1.2	9.8	4%
80–119	4.5	3.9	1.0	1.6	9.2	4%
120+	4.2	3.9	0.7	2.1	8.6	3%

H The four first moves

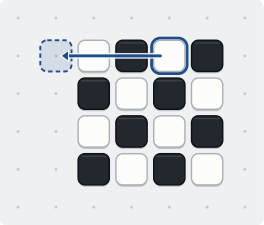
Up to symmetry, White has only four first moves, and none gives a clear advantage. The results come from 40 to 50 games between strong computer players per opening: differences of a few points are not significant. Opening theory remains to be written.



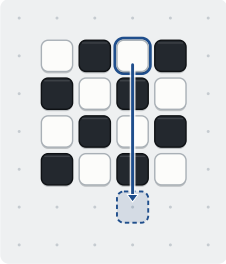
A · the corner piece slides along the edge. White 47%, Black 47%.



B · an edge piece slides outwards, along the edge. White 47%, Black 50%.



C · the same piece slides the other way, over two pieces. White 51%, Black 47%.



D · the same piece crosses the square from top to bottom. White 28%, Black 60% when imposed; 46% / 46% when the computer chooses it freely.